FIBRE GEOMETRY OF A CUT-DEPENDENT CROSSING-COMPONENT RETRACTION

ANONYMOUS

Abstract. Fix a cut in a chord matching and, simultaneously for every component of its crossing graph, replace the component by consecutive pairs on the same endpoint support. This is a one-step retraction onto noncrossing matchings. For a target matching T, its nesting forest turns the inverse problem into independent choices on the ordered lists of immediate siblings. We prove an all-size bijection: partition each sibling list noncrossingly and decorate every part by a crossing-connected matching. It gives the pointwise fibre product

$$|\Phi_n^{-1}(T)| = \prod_v a_{d_T(v)},$$

where $d_T(v)$ is a child degree, including the virtual root, and a_d is the already-known noncrossing-partition transform of connected chord diagrams. Strict supermultiplicativity then identifies the consecutive matching as the unique largest-fibre target. Component decompositions, parallel parts, Catalan counts, the transform and its coefficient sequence are explicitly treated as background. The owner search is bounded; no novelty or priority claim is made, and external release remains on hold.

1. The rooted map and the subtraction boundary

Put the endpoints $1 < \cdots < 2n$ on a circle with a fixed cut immediately before 1. Let $\mathcal{M}_n$ be the perfect matchings of these endpoints. Thus "rooted" means the fixed linear order, not a distinguished chord. Chords (a, b) and (c, d), written with smaller endpoint first, *cross* when $a < c < b < d$ or $c < a < d < b$. The crossing graph $G(M)$ has the chords of M as vertices and these crossings as edges.

For a component K of $G(M)$, sort all its endpoints as $s_1 < \cdots < s_{2k}$. Define $\Phi_n(M)$ by replacing, simultaneously for every component, its chords by

$$(s_1, s_2), (s_3, s_4), \ldots, (s_{2k-1}, s_{2k}). \tag{1}$$

The cut is essential: rotating an unrooted diagram need not commute with (1).

We record the ownership boundary before proving anything. Kreweras owns the classical noncrossing-partition framework [6]; Flajolet–Noy own connected chord-diagram decomposition and enumeration [4]; and Nabergall gives a decorated even-block decomposition in this setting [8]. Acan treats chord intersection graphs and their components [1]. Callan owns the generic noncrossing-partition transform [3], while Alman–Lian–Tran directly enumerate the full-wiring sequence that occurs below, including its recurrence and asymptotics [2]. Igusa's definition of parallel parts and his adjacency criterion [5, Definition 1.7 and Proposition 1.8] specialize exactly to the nonempty matching sibling lists used below: top-level chords are maximal parts, while members of a nonempty list of immediate siblings are covered by the same parent chord and no endpoint of that parent lies between them. Degree-zero lists contribute only the singleton $\mathcal{A}_0$ bookkeeping factor. Sibling localization and compatible noncrossing merging therefore receive zero credit, as does Thomas Lam's uncrossing order on matchings [7]. In particular, the Lam paper is by *Thomas Lam*, not by Alman–Lian–Tran. We claim only the literal cut-dependent section together with its connected decorations, target-wise fibre product and unique maximizer. The bounded search found no direct owner of that conjunction; a bounded non-hit is not a novelty certificate.

For a matching M, let $\pi(M)$ be the partition of $[2n]$ into endpoint supports of the components of $G(M)$. The classical component-support lemma says that $\pi(M)$ is a noncrossing partition into even blocks [4, 8]. If P is such a partition, let $s(P)$ pair successive elements of each block. Then

$$\Phi_n = s \circ \pi. \tag{2}$$

Proposition 1.1 (background dynamics)**.** *The map Φ_n is a retraction onto the noncrossing matchings:*

$$\Phi_n^2 = \Phi_n, \qquad \operatorname{im} \Phi_n = \operatorname{Fix} \Phi_n = \mathrm{NC}_2(n).$$

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Hence the image and fixed-set size is Cat_n*, every nonfixed state has depth one and indegree zero, and*

$$\zeta_{\Phi_n}(z) = (1-z)^{-\mathrm{Cat}_n}.$$

For $n = 0$*,* $\mathcal{M}_0 = \{\varnothing\}$*,* $(-1)!! = \mathrm{Cat}_0 = 1$*, and the empty matching is fixed.*

Proof. Equation (2) makes every image noncrossing. A noncrossing matching has singleton crossing components and is unchanged, proving the retraction statements and the Catalan count. If $\Phi(y) = x$ then $\Phi(x) = \Phi^2(y) = x$, so a nonfixed x has no predecessor. Finally every positive iterate has the same Cat_n fixed points, which gives the stated fixed-n Artin–Mazur zeta function. These are formal consequences of a one-step retraction and receive no contribution credit. □

2. The section fibre and its sibling inverse

Fix $T \in \mathrm{NC}_2(n)$. Its *nesting forest* has one vertex per chord: a chord $x = (a, b)$ is below the least chord (c, d) satisfying $c < a < b < d$, if one exists. Add a virtual root $\widehat{0}$ above all top-level chords. Children are ordered from left to right. Write $d_T(v)$ for the number of immediate children of v, including $v = \widehat{0}$.

Let c_k be the number of crossing-connected matchings on $[2k]$ and put $c_1 = 1$. Define $\mathcal{A}_d$ to consist of a noncrossing partition $\rho \in \mathrm{NC}(d)$ together with, for every block B of ρ, a crossing-connected matching on $[2|B|]$. Set

$$a_d = |\mathcal{A}_d| = \sum_{\rho \in \mathrm{NC}(d)} \prod_{B \in \rho} c_{|B|}, \qquad a_0 = 1. \tag{3}$$

We use one elementary property repeatedly: if P is a noncrossing partition and B is one of its blocks, every other block lies in one of the open cyclic gaps between consecutive elements of B. This is equivalent to the absence of alternating elements from two blocks.

Theorem 2.1 (all-size sibling inverse)**.** *For every* $T \in \mathrm{NC}_2(n)$*, extraction from a source gives a bijection*

$$\Phi_n^{-1}(T) \longleftrightarrow \prod_{v \in V(T) \cup \{\widehat{0}\}} \mathcal{A}_{d_T(v)}. \tag{4}$$

Consequently

$$|\Phi_n^{-1}(T)| = \prod_{v \in V(T) \cup \{\widehat{0}\}} a_{d_T(v)}. \tag{5}$$

Proof. We prove the promised inverse in four directions.

Step 1: forward localization. Take $M \in \Phi_n^{-1}(T)$ and set $P = \pi(M)$. For a block $B = \{b_1 < \cdots < b_{2k}\}$ of P, its section chords are $x_i = (b_{2i-1}, b_{2i})$. They are pairwise disjoint, so no two are in an ancestor–descendant relation. We claim that they are immediate siblings. If x_i has parent p_i, the endpoints of p_i belong to a different block of P. By the cyclic-gap property, those endpoints can enclose x_i only by lying on opposite sides in the outer gap of B; hence p_i encloses *every* element of B. Thus the parents of any two x_i and x_j both enclose all section chords. They are comparable by nesting, and if they were unequal, say p_i lay strictly inside p_j, then p_i would enclose x_j as well. Thus p_i would be a strict intermediate container between x_j and its alleged immediate parent p_j, a contradiction. Hence all parents coincide. If one x_i is top-level, any parent of another x_j would enclose x_i as well, again a contradiction. The claim follows, with the virtual root in the top-level case.

For a fixed parent v, the P-blocks therefore group its ordered sibling list. These groups form a noncrossing partition of the sibling indices. Indeed, groups containing indices $i < k$ and $j < \ell$, with $i < j < k < \ell$, would give alternating endpoints from their two P-blocks, contrary to noncrossing of P. We have extracted a unique $\rho_v \in \mathrm{NC}(d_T(v))$ for every v.

Step 2: converse section construction. Conversely choose all ρ_v independently. For each block Q of ρ_v, form an endpoint block B_Q by taking both endpoints of the siblings indexed by Q. Every target chord is a child exactly once and every sibling index belongs to one block of its ρ_v; hence the B_Q are disjoint and cover $[2n]$. They therefore form an even-block partition P.

We prove that P is noncrossing by induction from the leaves. Assume for a parent v that the blocks constructed strictly below each child already form a noncrossing partition inside that child's open interval. At v, nonalternation of the index blocks in ρ_v is exactly nonalternation of the corresponding endpoint blocks. Fix a child $x = x_i = (a, b)$ and a parent-level block B_Q. If $i \in Q$, then a, b are consecutive in the sorted list of B_Q, so the descendant interval $I_x = (a, b)$ is a gap of B_Q. If $i \notin Q$, the whole closed support $[a, b]$ lies strictly inside one gap of B_Q (the outer cyclic gap when all members of Q lie on one side). Thus every block constructed strictly below x, being contained in I_x, lies in one gap of every parent-level block.

Blocks below different children lie in disjoint intervals, and the already distinct B_Q neither merge nor alternate across levels. The leaf induction therefore proves noncrossing in every chord subtree. The same two-case argument for the top-level children at the virtual root closes the induction and proves that no two blocks of P cross.

If $Q = \{i_1 < \cdots < i_r\}$, the elements of B_Q occur as the left and right endpoints of child i_1, then those of child i_2, and so on. Descendant endpoints lie inside these chord intervals but are in other blocks. Successive pairing inside B_Q therefore recovers precisely those siblings. Every chord is a child exactly once, so $s(P) = T$.

Step 3: connected decorations. For each Q, choose the connected matching recorded by its $\mathcal{A}$-decoration and transport it order-preservingly from $[2|Q|]$ to B_Q. Take the union over all Q and all parents. Chords on distinct blocks cannot cross because P is noncrossing; within B_Q the decoration is crossing-connected. Thus the crossing components of the constructed source M have supports *exactly* the blocks B_Q. It follows that $\pi(M) = P$ and $\Phi_n(M) = s(P) = T$.

Step 4: mutual inverse. Starting from M, its crossing components recover P uniquely. Step 1 recovers the unique sibling groups, and order-standardizing each component recovers its connected decoration. Step 3 then returns the same chords of M. In the other direction, Step 3 proves that the constructed blocks are the exact crossing components, so extraction returns every chosen ρ_v and decoration. The constructions are mutual inverses in every size, including repeated block sizes and the virtual-root case. Independence over v now gives (4) and (5). □

Remark 2.2 (firewall). For the noncrossing endpoint partition whose parts are the chords of T, each nonempty immediate-sibling list (including the nonempty top-level list) is exactly an Igusa parallel set [5, Definition 1.7]; his [5, Proposition 1.8] also owns the compatible-merge criterion. An empty child list contributes only the singleton factor $\mathcal{A}_0$ and is not identified with an Igusa parallel set. The localization and static merge geometry in Steps 1–2 are therefore fully zero-credit. The residual is only their use inside this specified cut-dependent section, where connected decorations identify the exact crossing components and yield every target fibre. The nesting forest is a proof coordinate, not a tree dynamic. This separates the result from the cyclic partition shift–join dynamics of P110 (whose deepest-shell witness uses one two-element chord block), the tree involution of P120, graph-component complementation in P123, and the run/composition maps P117, P122 and P126. Componentwise behavior alone receives no portfolio credit.

3. Formal transform and the unique largest fibre

Let

$$C(u) = \sum_{k\geq 1} c_k u^k, \qquad A(u) = \sum_{d\geq 0} a_d u^d.$$

The owned noncrossing-partition transform applied to (3) gives the identity of *formal* ordinary power series

$$A(u) = 1 + C\big(uA(u)\big). \tag{6}$$

There is no convergence or asymptotic claim: connected chord counts grow factorially. The first values are

d	0	1	2	3	4	5	6	7
c_d	–	1	1	4	27	248	2830	38232
a_d	1	1	2	8	52	464	5184	68928

This is an all-size identification, not an inference from the displayed prefix: the owned decomposition of full wiring diagrams into connected pieces underlying (6) identifies $\mathcal{A}_d$ with that class for every d. Thus $a_d = X_d$ in the notation of Alman–Lian–Tran. Their Theorem 4.1.6 gives the full-wiring recurrence, Remark 4.1.7 identifies the sequence A111088, Theorem 4.1.8 gives its coefficient identity, and Theorem 4.2.1 gives its asymptotic [2]. All of those enumerative facts, as well as the transform itself, receive zero credit here.

Theorem 3.1 (unique largest fibre)**.** *For every $n \geq 0$, the unique largest fibre of Φ_n lies over the consecutive matching*

$$T_n^{\mathrm{con}} = (1,2)(3,4)\cdots(2n-1,2n),$$

and has size a_n.

Proof. For $i, j > 0$, juxtaposition gives an injection $\mathcal{A}_i \times \mathcal{A}_j \hookrightarrow \mathcal{A}_{i+j}$. It is not onto: a one-block noncrossing partition with a connected $(i+j)$-chord decoration is outside its image. Such a decoration exists in every size $k = i + j$: pair r with $r + k$ for $1 \le r \le k$, making all chords cross. Therefore

$$a_i a_j < a_{i+j}. \tag{7}$$

Every chord has exactly one parent in the virtual-rooted forest, so $\sum_v d_T(v) = n$. Repeated use of (7), with $a_0 = 1$, shows that the product in (5) is at most a_n, strictly unless only one vertex has positive child degree. A chord vertex cannot have all n chords as children, since it would be an additional chord; hence that sole vertex must be $\widehat{0}$. All chords are then top-level. A noncrossing chord spanning a nonempty interval would contain another chord, so every chord is consecutive and $T = T_n^{\mathrm{con}}$. Conversely this target has $d_T(\widehat{0}) = n$ and all other degrees zero, giving fibre size a_n. The empty case has the sole empty target and uses $a_0 = 1$. □

Corollary 3.2 (background census)**.** *For every $n \ge 0$,*

$$\sum_{T \in \mathrm{NC}_2(n)} \prod_v a_{d_T(v)} = (2n-1)!!.$$

The consecutive target has fibre a_n, whereas the fully nested rainbow target has fibre 1; the fibres are already nonuniform at $n = 2$.

Proof. The sum partitions the finite domain $\mathcal{M}_n$ into the fibres of theorem 2.1. For the rainbow target every positive child degree is one and $a_1 = 1$. □

4. Exact finite control and claim ceiling

A paper-local standard-library verifier exhausts all matchings through seven chords: 146,600 states including the empty state and 626 noncrossing targets. It checks the literal map, component-support noncrossing, idempotence, forward sibling extraction, (5), the formal coefficients, strict inequalities, and an independent converse construction that rebuilds every one of the 146,600 sources. The canonical run makes 735,609 assertions. This is counterexample pressure, not a replacement for the all-size proof above.

The admissible contribution ceiling is exactly theorems 2.1 and 3.1 for the specified cut-dependent map. The Catalan image, idempotence, zeta, component counts, A111088, (6), generic parallel-part geometry, and generic uncrossing are background. No analytic asymptotic, unrooted canonicity, novelty, priority, or external-release assertion is made.

References

[1] Hüseyin Acan. On a uniformly random chord diagram and its intersection graph. *Discrete Mathematics*, 340:1967–1985, 2017. doi: 10.1016/j.disc.2016.11.004.

[2] Joshua Alman, Carl Lian, and Brandon Tran. Circular planar electrical networks: Posets and positivity. *Journal of Combinatorial Theory, Series A*, 132:58–101, 2015. doi: 10.1016/j.jcta.2014.11.004.

[3] David Callan. Sets, lists and noncrossing partitions. *Journal of Integer Sequences*, 11:Article 08.1.3, 2008. URL https://cs.uwaterloo.ca/journals/JIS/VOL11/Callan/callan412.html.

[4] Philippe Flajolet and Marc Noy. Analytic combinatorics of chord diagrams. In *Formal Power Series and Algebraic Combinatorics*, pages 191–201. Springer, Berlin, 2000. doi: 10.1007/978-3-662-04166-6_17.

[5] Kiyoshi Igusa. A category of noncrossing partitions. *Applied Categorical Structures*, 33(6):Article 40, 2025. doi: 10.1007/s10485-025-09838-8.

[6] Germain Kreweras. Sur les partitions non croisées d'un cycle. *Discrete Mathematics*, 1:333–350, 1972. doi: 10.1016/0012-365X(72)90041-6.

[7] Thomas Lam. The uncrossing partial order on matchings is eulerian. *Journal of Combinatorial Theory, Series A*, 135:105–111, 2015. doi: 10.1016/j.jcta.2015.04.004.

[8] Lukas Nabergall. *Enumerative Perspectives on Chord Diagrams*. PhD thesis, University of Waterloo, 2022. URL https://uwspace.uwaterloo.ca/items/51239c85-b044-4e6b-97c6-710332c37c93.